GEOGRAPHY

INDIAN GEOGRAPHY





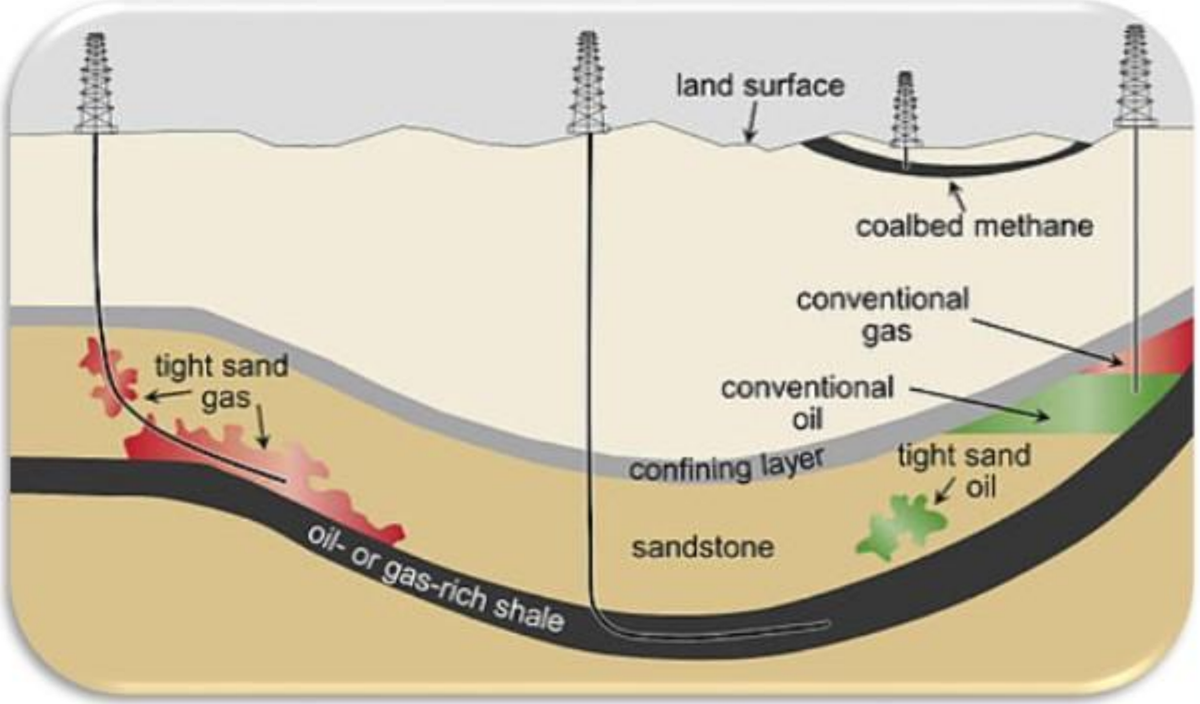
MINERALS & INDUSTRIES

MODULE - 6

- COAL BED METHANE
- SHALE GAS RESERVOIRS

INDIAN MINERAL & ENERGY RESOURCES

UNCONVENTIONAL GAS RESERVOIRS



1. Conventional reservoirs of oil and natural gas are found in permeable sandstone.

2. Unconventional Gas Reservoirs occur in relatively impermeable sandstones, in joints and fractures or absorbed into the matrix of shales and in coal.

3. Given current economic conditions and state of technology, they are more expensive to exploit.
Example: Shale gas, and coalbed methane.

4. Considerable quantities of methane is trapped within coal seams.

5. A significant portion of this gas remains as free gas in the joints and fractures of the coal seam.

INDIAN MINERAL & ENERGY RESOURCES

UNCONVENTIONAL GAS RESERVOIRS

6 This gas can be accessed by drilling wells into the coal seam and pumping large quantities of water that saturate the seam. [water will occupy the gaps and pores and will push out the gas]

7 It is now becoming an important source of energy.

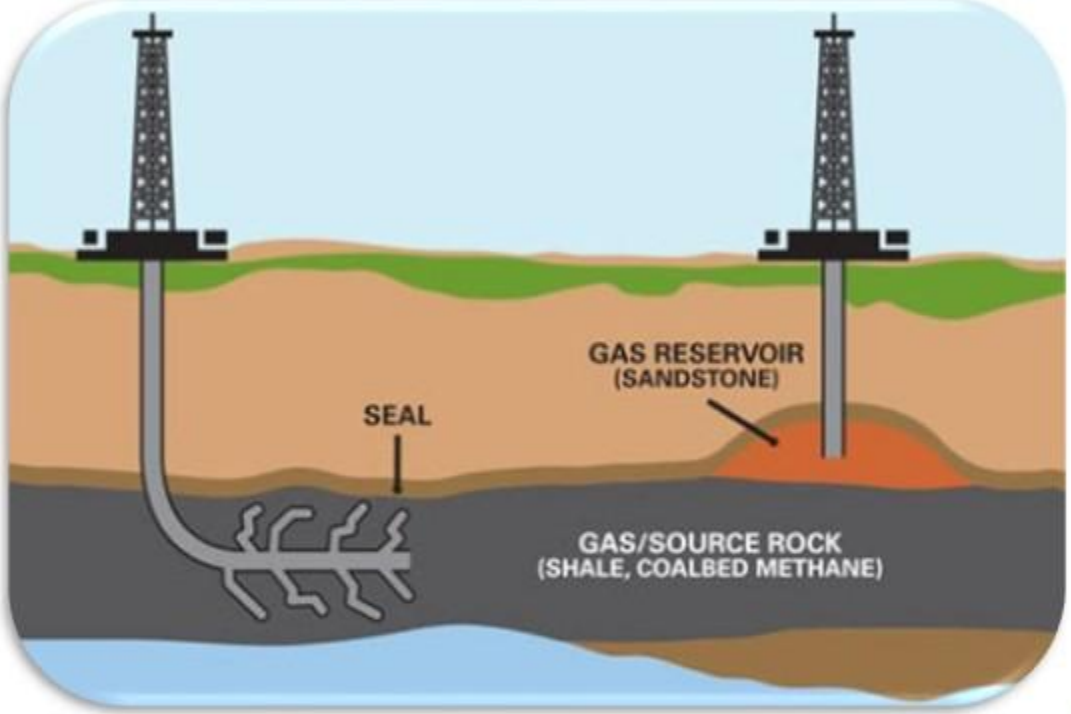
8 Unlike much natural gas from conventional reservoirs, coalbed methane contains very little heavier hydrocarbons such as propane or butane.

9 The presence of this gas is well known from its occurrence in underground coal mining, where it presents a serious safety risk.

10 Fire Accidents in Coal Mines are mainly due to Coalbed Methane, and Lignite deposits which undergo spontaneous combustion.

INDIAN MINERAL & ENERGY RESOURCES

COALBED METHANE IN INDIA



1. With one of the largest proven coal reserves, and one of the largest coal producer in the world, India holds significant prospects for commercial recovery of coalbed methane.
2. The country has an estimated 700-950 billion cubic metre of coalbed methane.

INDIAN MINERAL & ENERGY RESOURCES

COALBED METHANE IN INDIA

PROBLEMS IN EXPLORATION, EXTRACTION OF COALBED METHANE IN INDIA

CAPITAL INTENSIVE PROCESS

1. Extracting unconventional gas is a capital intensive process and at the present levels of gas prices, the companies cannot recover their investments.

TECHNOLOGY REQUIREMENTS

2. The technology required is very advanced and the public sector companies have very weak organizational setup to efficiently handle such technologies and extract gas economically.

INDIAN MINERAL & ENERGY RESOURCES

COALBED METHANE IN INDIA

PROBLEMS IN EXPLORATION, EXTRACTION OF COALBED METHANE IN INDIA

GAS PRICING

3 In India, gas pricing is a contentious issue. It has never been easy satisfying all the stakeholders involved [consumer, government, gas companies].

4 Gas pricing will be critical for private companies before they can invest in unconventional gas projects so that they can calculate their profit margin.

COALBED METHANE IN INDIA

PROBLEMS IN EXPLORATION, EXTRACTION OF COALBED METHANE IN INDIA

RIGHTS OF COAL BLOCK ALLOTTEES

5 There may be an overlap between companies having rights over coal and CBM.

6 Multiple ownership for simultaneous exploitation may not be desirable for the life, health and safety of the workers employed in such mines.

CONFLICT WITHIN MINISTRIES

7
- CBM extraction falls under Ministry of Petroleum & Natural Gas whereas coal mining falls under Ministry of Coal.
- This has resulted in the conflict [common feature of Indian Bureaucracy] between the two ministries and other associated bureaucratic hurdles.

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COALBED METHANE IN INDIA

PROBLEMS IN EXPLORATION, EXTRACTION OF COALBED METHANE IN INDIA

LAND ACQUISITION AND WATER HANDLING PROBLEMS.

8 Land acquisition problems for CBM blocks are severe.

9 Also some blocks are located in the tribal land areas of Jharkhand. Chhattisgarh etc.

10 CBM wells tend to produce large volumes of water during the initial period which gets accumulated in the upper strata of the coal seams.

11 Disposing this water which has high salinity constitutes a big problem.

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COALBED METHANE IN INDIA

PROBLEMS IN EXPLORATION, EXTRACTION OF COALBED METHANE IN INDIA

AVAILABILITY OF GAS INFRASTRUCTURE

12 Many of the CBM gas fields do not have critical infrastructure like pipelines.

13 Also, many CBM blocks in the states like Jharkhand, Chhattisgarh, Madhya Pradesh do not have any near by industrial belt for local consumption of CBM.

DELAY IN GRANT OF PERMISSION FROM STATE GOVERNMENTS

DELAY IN GRANT OF ENVIRONMENTAL CLEARANCES

However, many of the above problems are being overcome by the recent governmental policies like Hydrocarbon Exploration Licensing Policy.

INDIAN MINERAL & ENERGY RESOURCES

SHALE GAS RESERVES

1. Shales are fine-grained sedimentary rocks formed of organic-rich mud at the bottom of ancient seas.

2. Subsequent sedimentation and the resultant heat and pressure transformed the mud into shale and also produced natural gas from the organic matter contained in it.

3. Over long spans of geologic time, some of the gas remained locked in the nonporous shale, forming shale gas accumulations.

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SHALE GAS RESERVES ACROSS THE WORLD

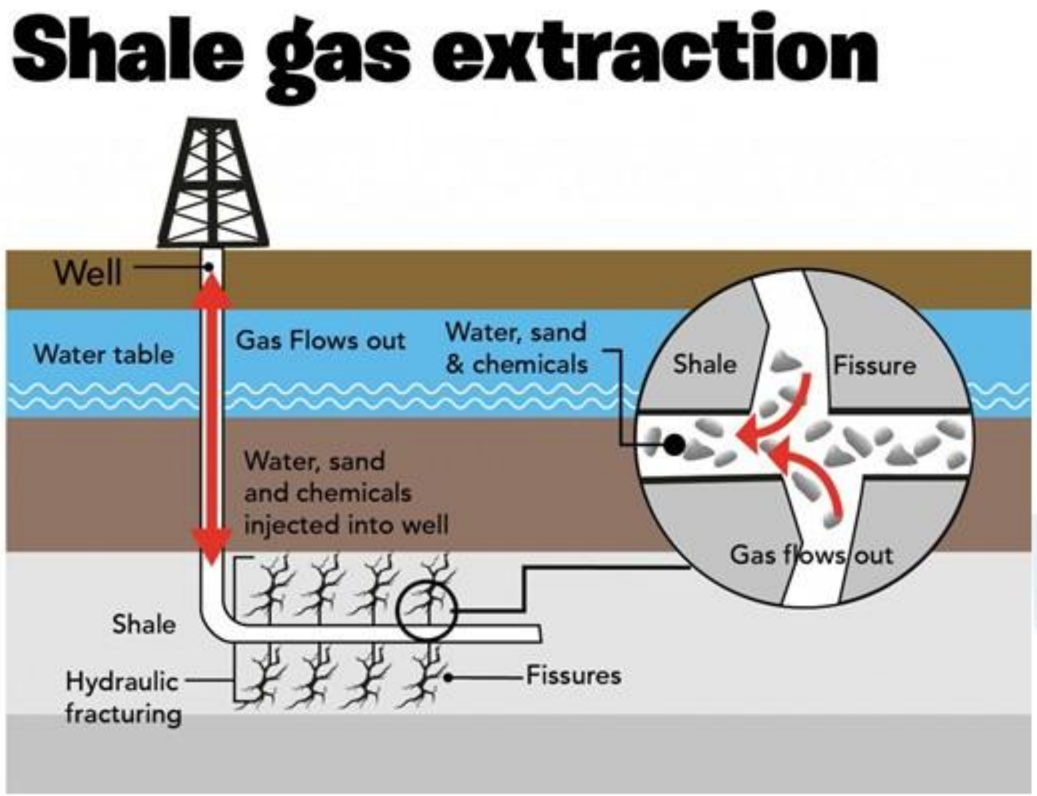
1. Basins of preliminary interest identified by Indian geologists are the Cambay Basin in Gujarat, the Assam-Arakan basin in northeast India, and the Gondwana Basin.

2. Indian engineers have gathered experience on fracking – the technology to find shale gas – by spending time in the US and are now able to hunt for the scarce resource on their own.

3. Fracking technology sends high pressure streams of water, sand and chemicals into shale formations to bring up the oil and gas.

4. Environmentalists have objected to fracking because of the damage to forest cover and possible contamination of ground water.

EXTRACTION OF SHALE GAS



1. Shale gas occurs frequently at depths exceeding 1,500 metres (5,000 feet).

2. The process of extracting shale oil and gas requires deep vertical drilling followed by horizontal drilling.

2. This is followed by hydraulic fracturing, or fracking, of the rock by the injecting of fluid at extremely high pressure.

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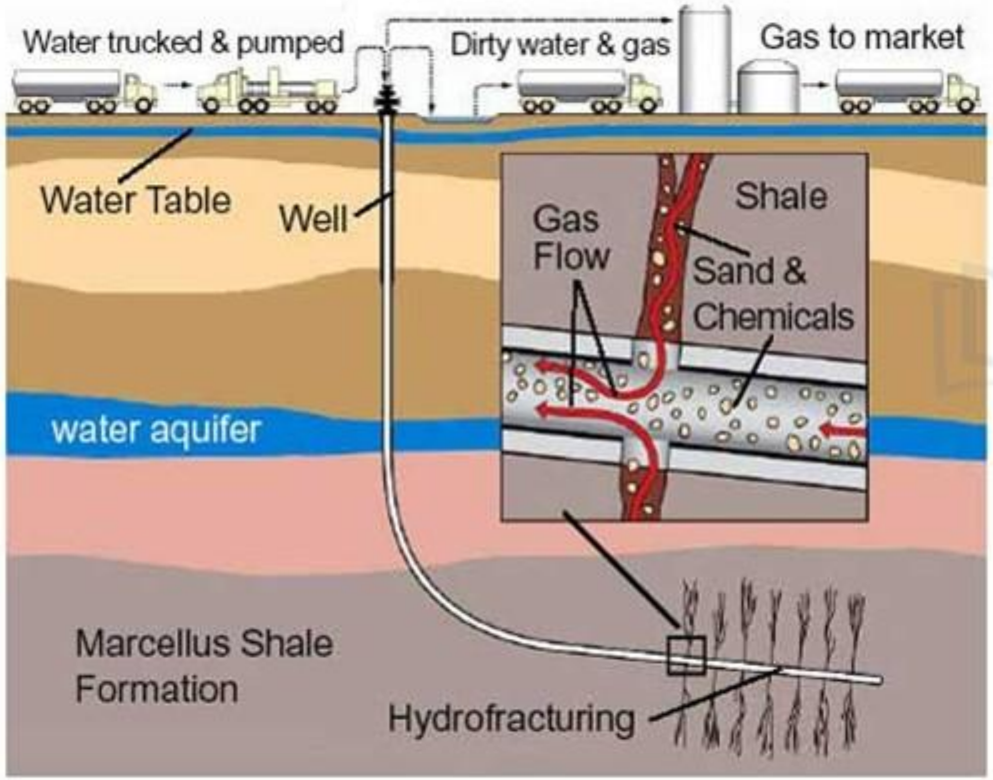
EXTRACTION OF SHALE GAS

WHAT DO YOU UNDERSTAND BY THE PROCESS OF HYDRO-FRACTURING OR FRACKING?

1. Shale rock is sometimes found 3,000 metres below the surface.



2. A mixture of water, chemicals, and sand is then injected into the well at very high pressures to create a number of fissures in the rock to release the gas.

3. The process of using water for breaking up the rock is known as 'hydro-fracturing' or 'fracking'.

4. The chemicals help in water and gas flow and tiny particles of sand enter the fissures to keep them open and allow the gas to flow to the surface.

Learning Space

INDIAN MINERAL & ENERGY RESOURCES

EXTRACTION OF SHALE GAS

WHAT IS THE ROLE PLAYED BY GUAR-GUM?

1. The guar bean is grown mainly by farmers in Rajasthan and Haryana.
2. It can quickly turn water into a very thick gel.
3. Adding guar gum increases viscosity of water and makes high-pressure pumping and the fracturing process more efficient.
4. High viscosity water is much more effective at suspending sand grains and carrying them into the fractures.
5. Earlier, guar gum was used mainly as an additive in ice creams and sauces.
6. But with the discovery of its use in shale gas extraction, its price shot up enormously.

INDIAN MINERAL & ENERGY RESOURCES

EXTRACTION OF SHALE GAS

Hydraulic Fracking Needs	INDIA'S PROBLEM
Large area	• land acquisition • environment clearance
Large water supply	• Shortage of water
Guar Gum (Fluid Viscosity Agent)	• Guar gum too costly. • India is largest producer of Guar gum. • But due to heavy imports from US shalegas companies, its prices have increased by ten times within a year. • Thus shortage for domestic industry who cannot keep up with the prices offered by Americans.

WHAT ARE THE PROBLEMS?

1. Environmentalists have objected to fracking because of the damage to forest cover and possible contamination of ground water.
2. The process requires around 5 to 9 million litres of water per extraction activity. It poses a daunting challenge to India's fresh water resources.
3. However, industry officials say that the treated water can be re-used for further fracking and need not be disposed of at all.

INDIAN MINERAL & ENERGY RESOURCES

EXTRACTION OF SHALE GAS

IF U.S. CAN, THEN WHY CAN'T INDIA?

WATER SCARCITY	1	India suffers from water scarcity whereas the U.S. do not have the same water worries.
EXEMPTION FROM SCRUTINY	2	In the US, the natural gas department is exempt from scrutiny for chemical injection in the ground (it exempts companies from disclosing the chemicals used during hydraulic fracturing). There is no such legislation in India.
OWNERSHIP ISSUES	3	In US, the citizen or resident owns the resources that lie beneath the ground. In India, soil below the land is a public property and the companies must follow all the necessary rules to acquire it.

INDIAN MINERAL & ENERGY RESOURCES

EXTRACTION OF SHALE GAS

IF U.S. CAN, THEN WHY CAN'T INDIA?

MAPPING OF RESERVES → **4** The US has mapped all its shale reserves. In India there is clarity on the exact recoverable shale reserves.

POPULATION DENSITY → **5** The population density is much lower in the US and they can afford to do it.

LACK OF PIPELINES → **6** All locations in US is well connected with gas pipelines. Bulk of the reserves in eastern India lack the necessary network of pipelines to transport the gas–a task that many private operators are wary about undertaking.

INDIAN MINERAL & ENERGY RESOURCES

EXTRACTION OF SHALE GAS

FINAL VERDICT...

1. India has 293 billion tonnes of coal lying under its soil. Extraction is complicated because of environmental issues.
2. But, 'underground coal gasification', can create 6,900 trillion cubic feet of gas which is way HIGHER than shale reserves.
3. India's shale resources are at a more modest 65 trillion cubic feet. India's CBM potential is estimated at 450 tcf.
4. So, focus must be on CBM exploration rather than on risky shale business.

UPSC MAINS QUESTION

It is said that India has substantial reserves of shale oil and gas, which can feed the needs of the country. However, tapping of resources does not appear to be high on the agenda. Discuss critically the availability and issues involved.

(10 marks – 200 words)

INDIAN MINERAL & ENERGY RESOURCES

UNDERGROUND COAL GASIFICATION (UCG)

1. UCG is an in-situ gasification process carried out in non-mined coal seams using injection of oxidants, and bringing the product gas to surface through production wells drilled from the surface.

2. The predominant product gases are methane, hydrogen, carbon monoxide and carbon dioxide.

3. The technique can be applied to coal resources that are otherwise unprofitable or technically complicated to extract by traditional mining methods.

4. UCG offers an alternative to conventional coal mining methods for some resources.

5. Compared to traditional coal mining and processing, the underground coal gasification eliminates surface damage and solid waste discharge, and reduces sulfur dioxide (SO2) and nitrogen oxide (NOx) emissions.

6. UCG has the potential to exploit the coal resources regarded as either uneconomic to work by conventional underground coal extraction or in-accessible due to depth, geology or other mining and safety considerations.

INDIAN MINERAL & ENERGY RESOURCES

ECONOMIC BENEFITS OF UCG

1. No need to transport the coal.
2. No need for disposing of ash or slag.
3. No need for an above ground gasification plant.
4. No need to prepare the coal to be fed into a reactor.
5. No need for coal handling.
6. No need for the coal to be mined.
7. Significantly lower capital cost for project development than that of above ground plants.

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ENVIRONMENTAL BENEFITS OF UCG

1. Less environmental impacts traditionally associated with coal mining and handling.
2. All or a substantial portion of the sulfur, mercury, arsenic, tar, ash and particulates found in coal remain underground
3. Underground saline water is used.
4. Significantly reduced use of groundwater or freshwater.
5. No landfill disposal required for ash or slag.
6. Coal seams being gasified are below the fresh drinking water supplies.
7. Minimal land use.
8. Significantly reduced criteria air pollutants.
9. Any Sulphur or metals that reach the surface do so in a chemically reduced state, making them easier to remove.

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